

Week 04 Task 10

What is SIEM? + Install Splunk / Wazuh

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Table of Contents

1) Executive Summary 3
2) Introduction to SIEM 3
3) Working of SIEM 3
4) Installation of Wazuh SIEM Server 3
    4.1) Assigning a static IP to Wazuh server 5
5) Installation of Wazuh Agents 6
    5.1) Installation of Wazuh Agent on Windows 6
    5.2) Installation of Wazuh Agent on Linux 8
6) Conclusion 11

1) Executive Summary

This report presents an overview of Security Information and Event Management (SIEM), its role in centralized log management, threat detection, and incident response. It covers the working principles of SIEM solutions and introduces Wazuh as a platform for security monitoring and analysis.

The practical work involved installing and accessing the Wazuh SIEM platform, configuring a static IP address for the Ubuntu server, and deploying Wazuh agents on Windows 10 and Ubuntu endpoints. Through this exercise, I gained practical experience in configuring a centralized security monitoring environment and connecting endpoints for monitoring. This task strengthened my understanding of SIEM operations and the importance of centralized security monitoring in identifying and investigating potential security incidents.

2) Introduction to SIEM

SIEM stands for Security Information and Event Management. A Platform that provides a centralized dashboard for continuous monitoring across the organization's digital infrastructure. It collects and analyzes logs from the different sources such as firewalls, servers, network devices, DNS logs, EDR logs, Cloud logs and endpoints. It correlates events from these different sources to identify suspicious activities and potential security threats. The primary use cases of SIEM include centralized log management, real-time threat detection and alerting, and accelerated incident response and triage. There are many SIEM solutions such as **IBM QRadar**, **Splunk**, **Microsoft Sentinel** and **Wazuh**.

3) Working of SIEM

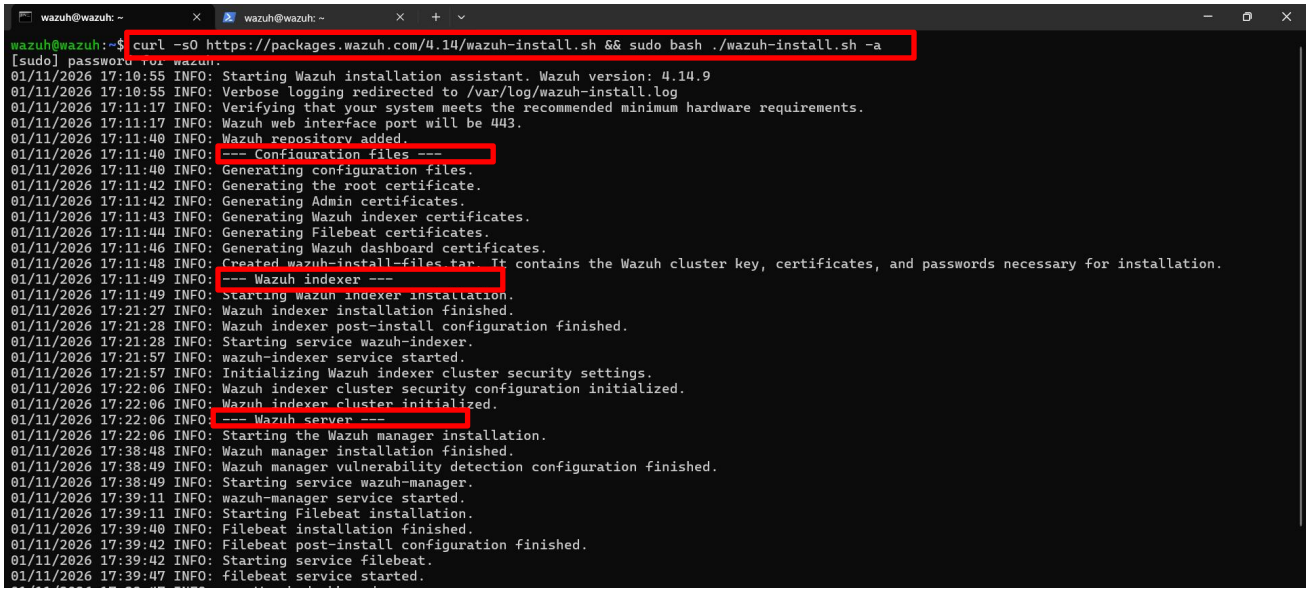
A Security Information and Event Management (SIEM) system operates by collecting, processing, analyzing, and correlating security-related data from multiple sources across an organization's IT infrastructure. These sources may include servers, endpoints, firewalls, network devices, and cloud services. The collected logs and events are centralized to provide security analysts with a unified view of the organization's security environment.

Once the data is collected, the SIEM system normalizes and analyzes the events using predefined rules, threat intelligence, and correlation techniques. It identifies unusual activities and patterns that may indicate security threats, such as repeated failed login attempts, unauthorized access, or suspicious network communication. When potentially malicious activity is detected, the system generates alerts that help SOC analysts investigate, prioritize, and respond to incidents. This process supports continuous security monitoring, threat detection, and incident response.

4) Installation of Wazuh SIEM Server

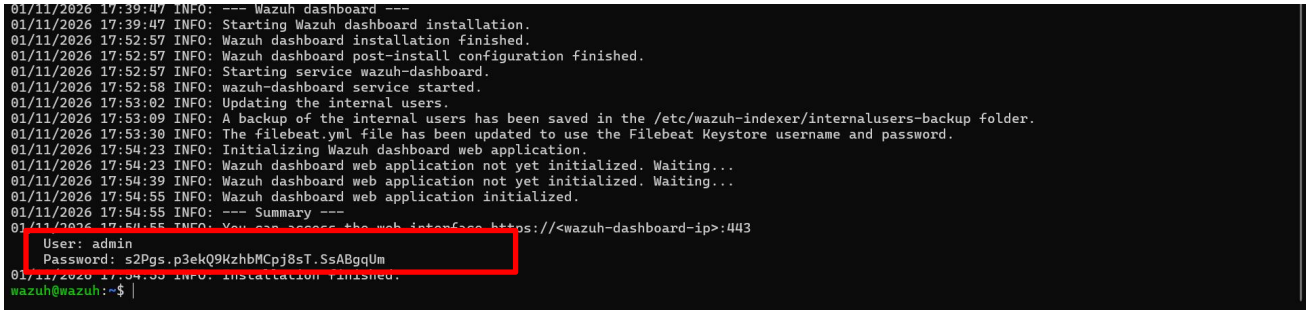
I have selected Wazuh SIEM because it's open source and free to use, supports both Windows and Linux endpoints, and includes built-in features such as file integrity monitoring and vulnerability detection. In order to download Wazuh SIEM go to official website of Wazuh and perform the steps mentioned below.

Official Wazuh Website Link: https://documentation.wazuh.com/current/quickstart.html

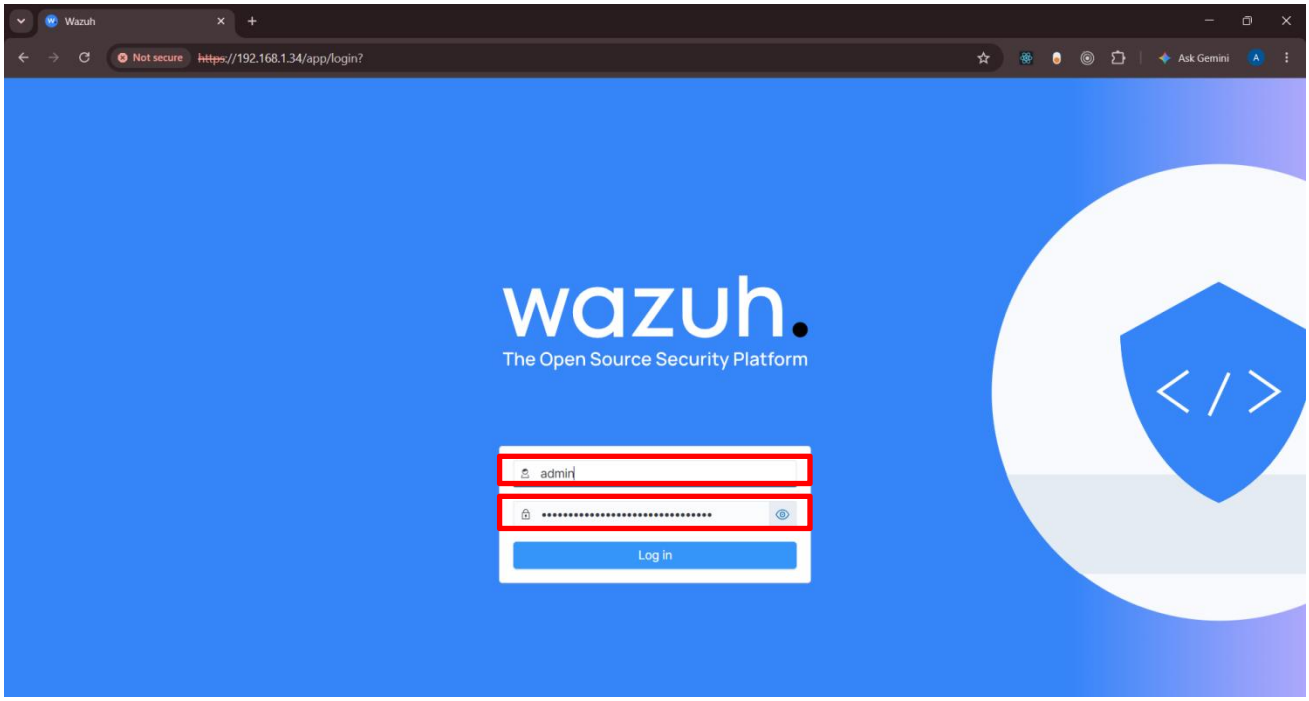


In the above figure a command is used to download the wazuh server. The installation deploys the Wazuh indexer, Wazuh server, and Wazuh dashboard, along with the required configuration files.

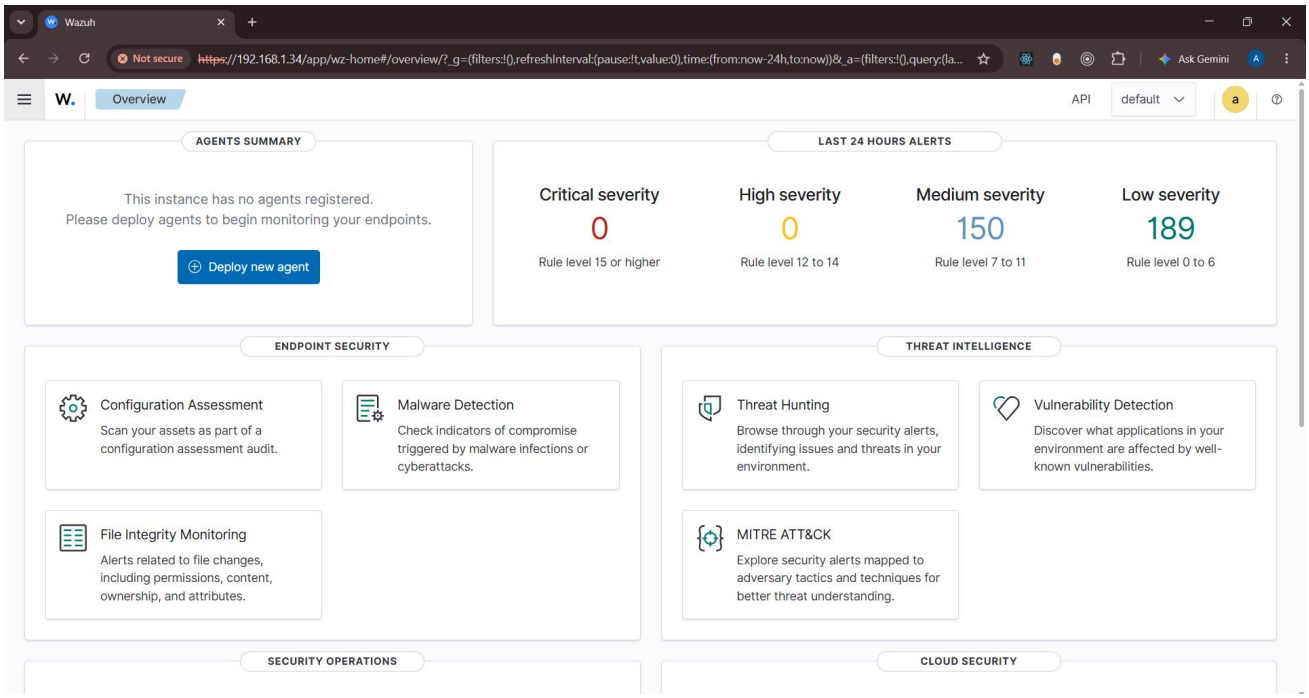
Command: **curl -sO https://packages.wazuh.com/4.14/wazuh-install.sh && sudo bash ./wazuh-install.sh -a**



The above screenshot displays the credentials generated during the Wazuh installation, indicating that the installation process has completed successfully.



The above screenshot shows the Wazuh dashboard after successful authentication, confirming that the web interface is accessible.



The above screenshot shows the Wazuh dashboard after successful authentication. The dashboard provides an overview of security alerts, endpoint security capabilities, and threat intelligence features, along with the option to deploy agents for endpoint monitoring.

4.1) Assigning a static IP to Wazuh server

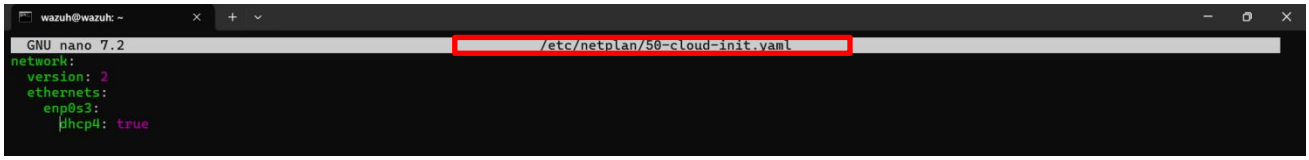
To maintain consistent communication between the Wazuh server and its agents, a static IP address is configured for the Ubuntu server. This prevents the server's address from changing due to DHCP lease assignments and avoids repeated updates to agent configurations.

Step 01



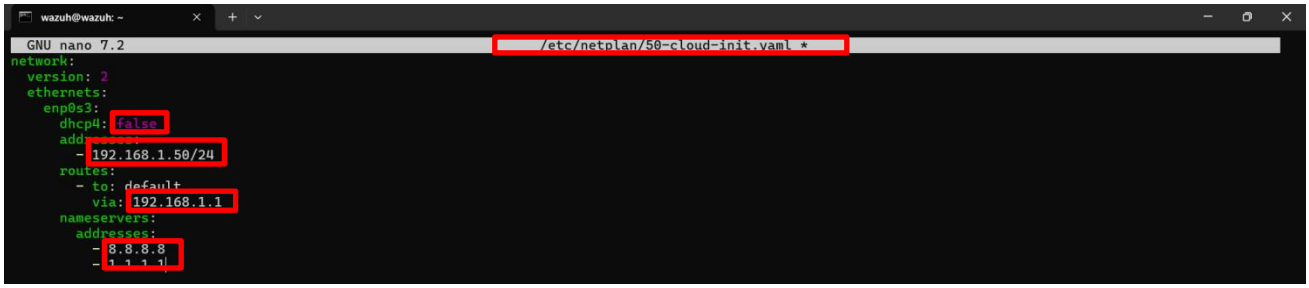
The above screenshot shows the Ubuntu network configuration file being accessed to configure a static IP address

Step 02



The above screenshot demonstrates the network configuration being updated with the static IP address, subnet, gateway, and DNS settings.

Step 03

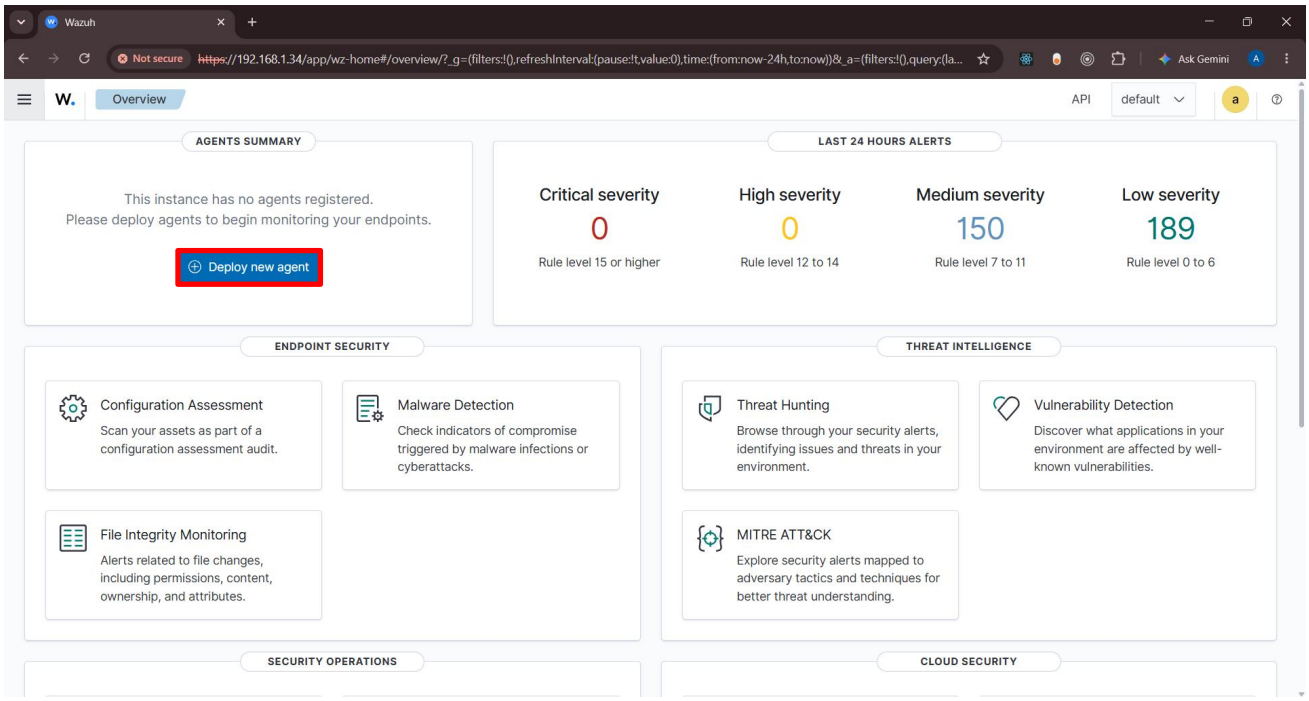


The above screenshot shows the application of the updated network configuration and verification of the assigned static IP address.

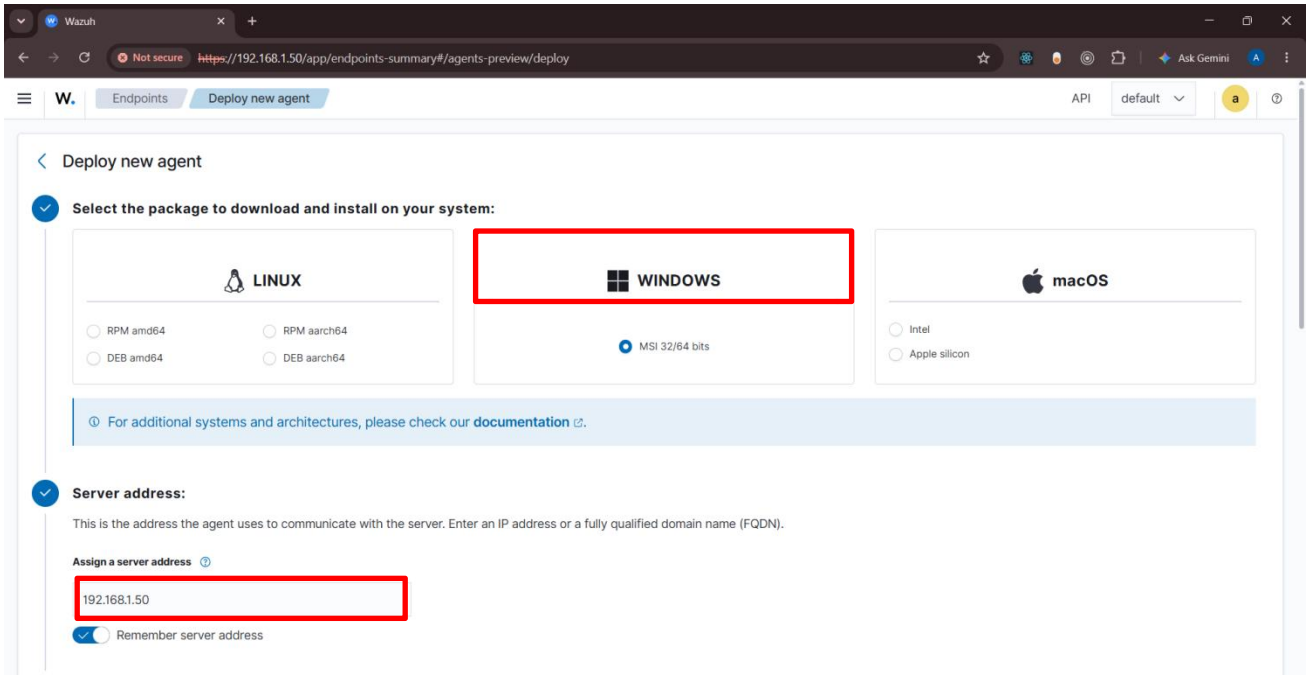
5) Installation of Wazuh Agents

The Wazuh agent is a multi-platform component of the Wazuh solution and runs on the endpoints we want to monitor. It communicates with the Wazuh server, sending data in near real-time through an encrypted and authenticated channel. The Wazuh agent provides capabilities such as log data collection, file integrity monitoring, threat detection, security configuration assessment, system inventory, vulnerability detection, and incident response to enhance your endpoint security.

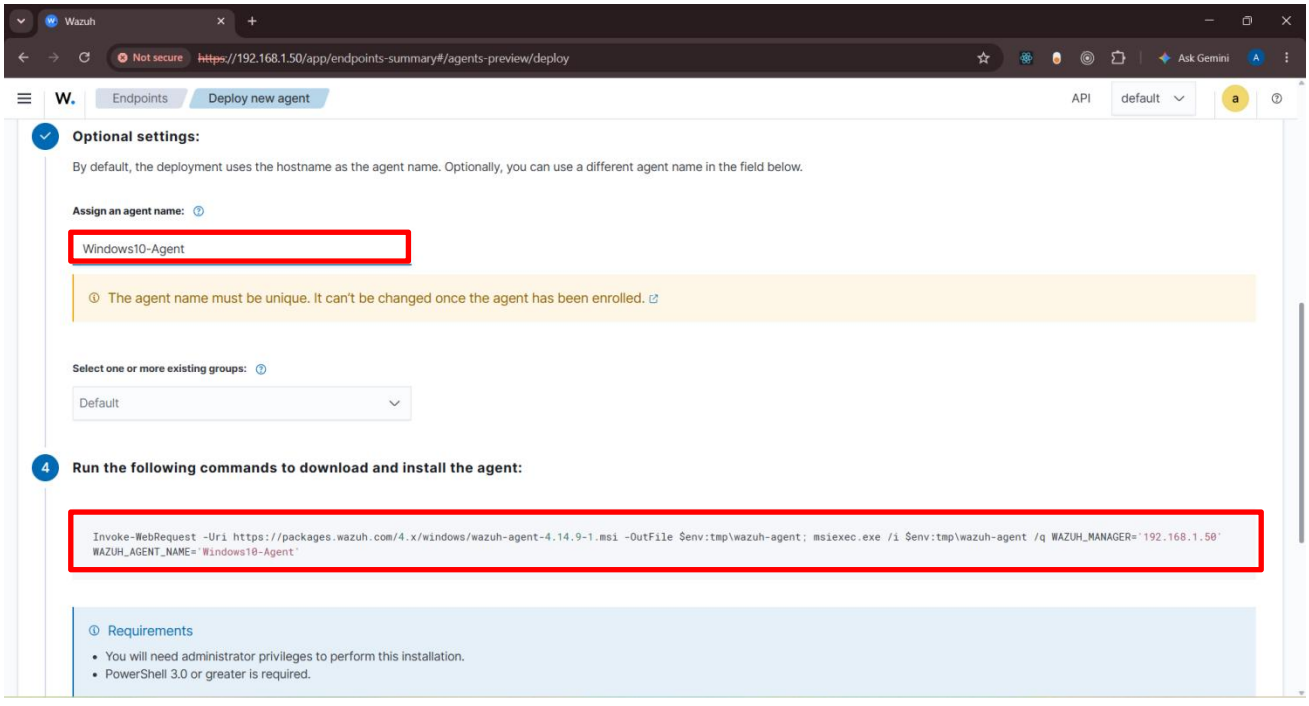
5.1) Installation of Wazuh Agent on Windows



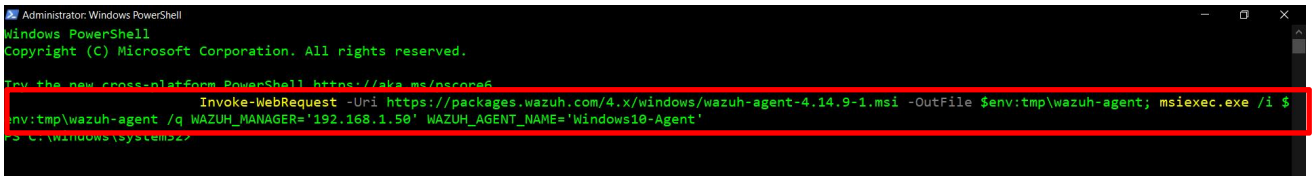
In the above figure Wazuh home page is shown where the Deploy new agent option is selected to begin configuring a monitored endpoint.



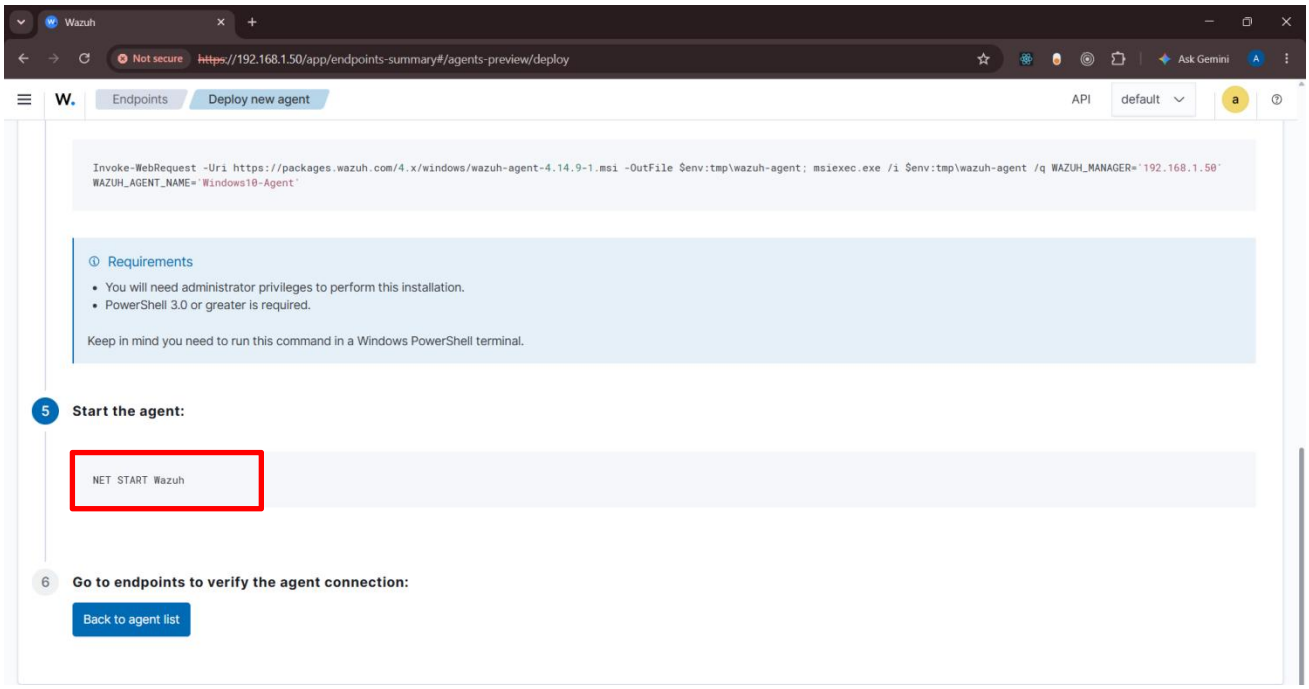
The above screenshot illustrates the Wazuh agent deployment configuration for a Windows endpoint. Windows is selected as the operating system, and the Wazuh server's IP address is specified to establish communication with the central monitoring server.



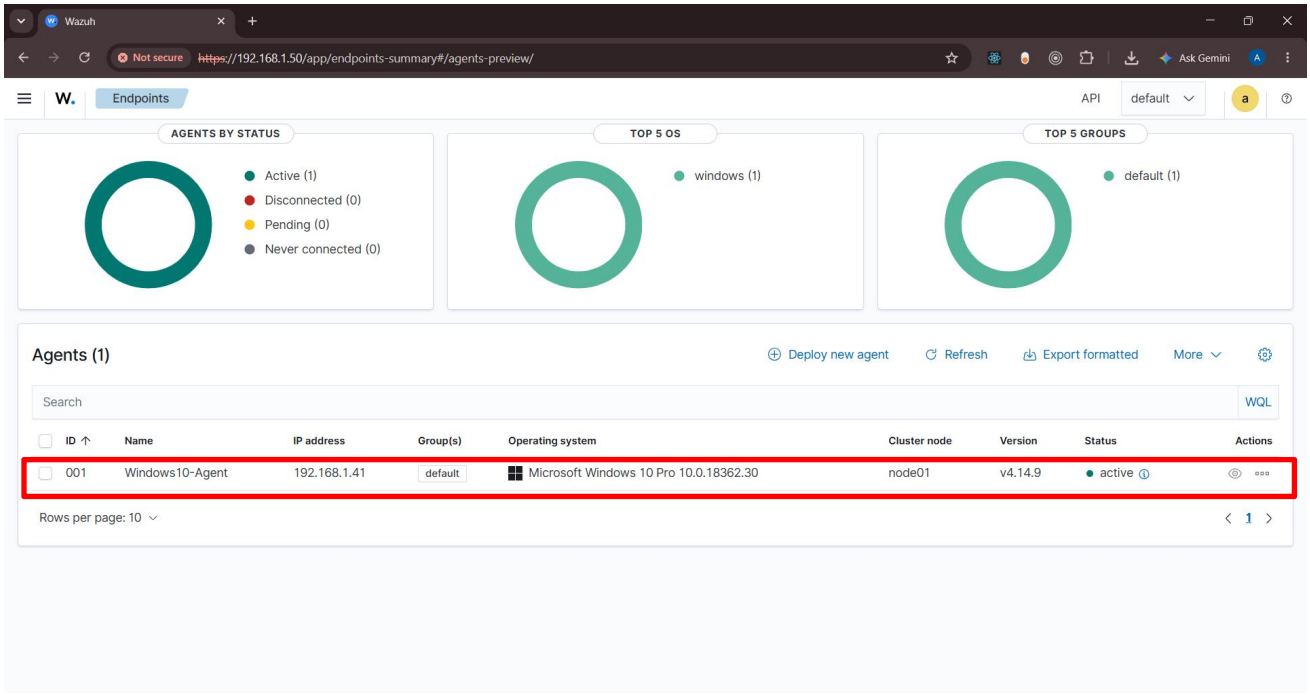
The above screenshot shows the Windows agent being identified as Windows10-Agent. The dashboard generates a PowerShell command highlighted to download and install the Wazuh agent on the endpoint.



The above screenshot demonstrates the execution of the Wazuh agent installation command in an elevated PowerShell session on Windows 10.

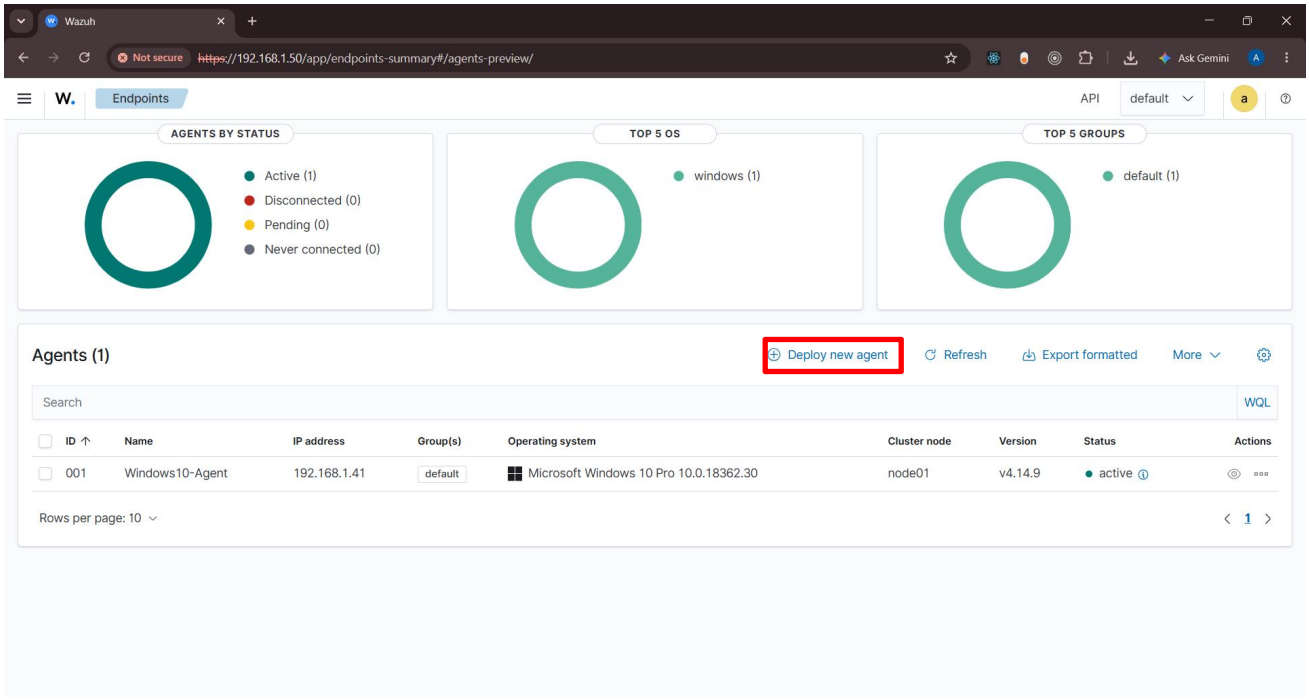


The above screenshot shows the command used to start the Wazuh agent service, enabling the endpoint to begin communicating with the Wazuh server.

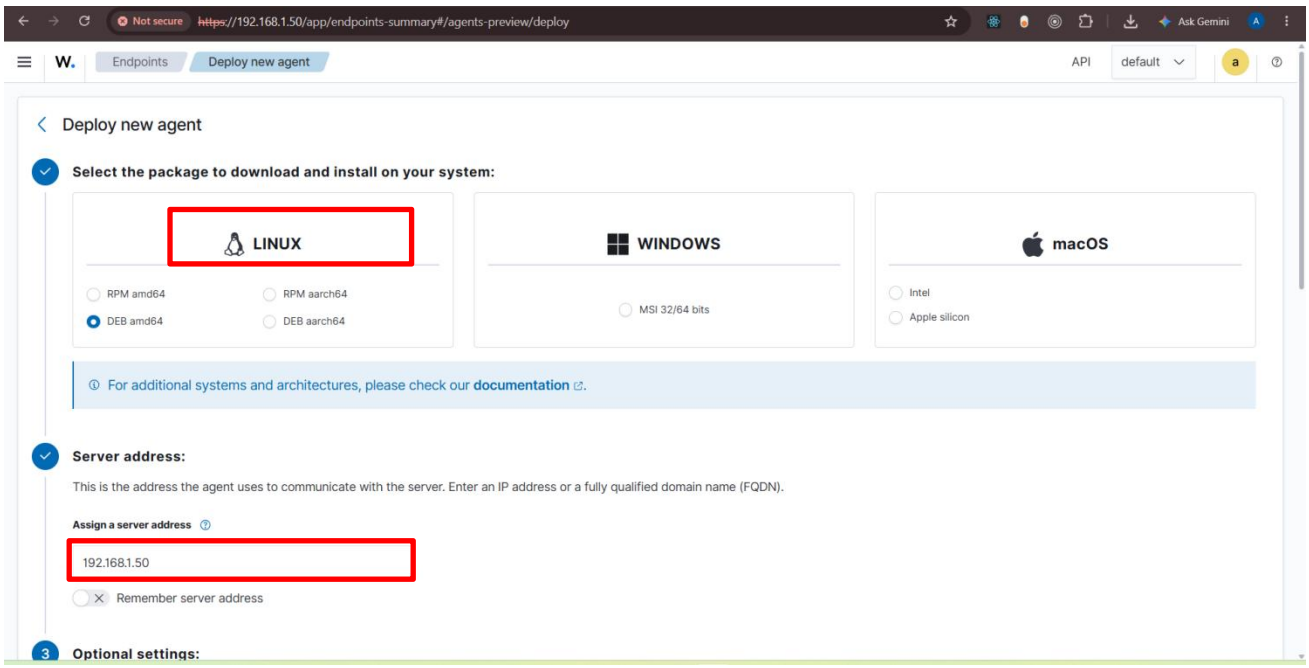


The above screenshot shows the Windows agent appearing in the Wazuh dashboard after installation, confirming that the endpoint has been registered with the monitoring server.

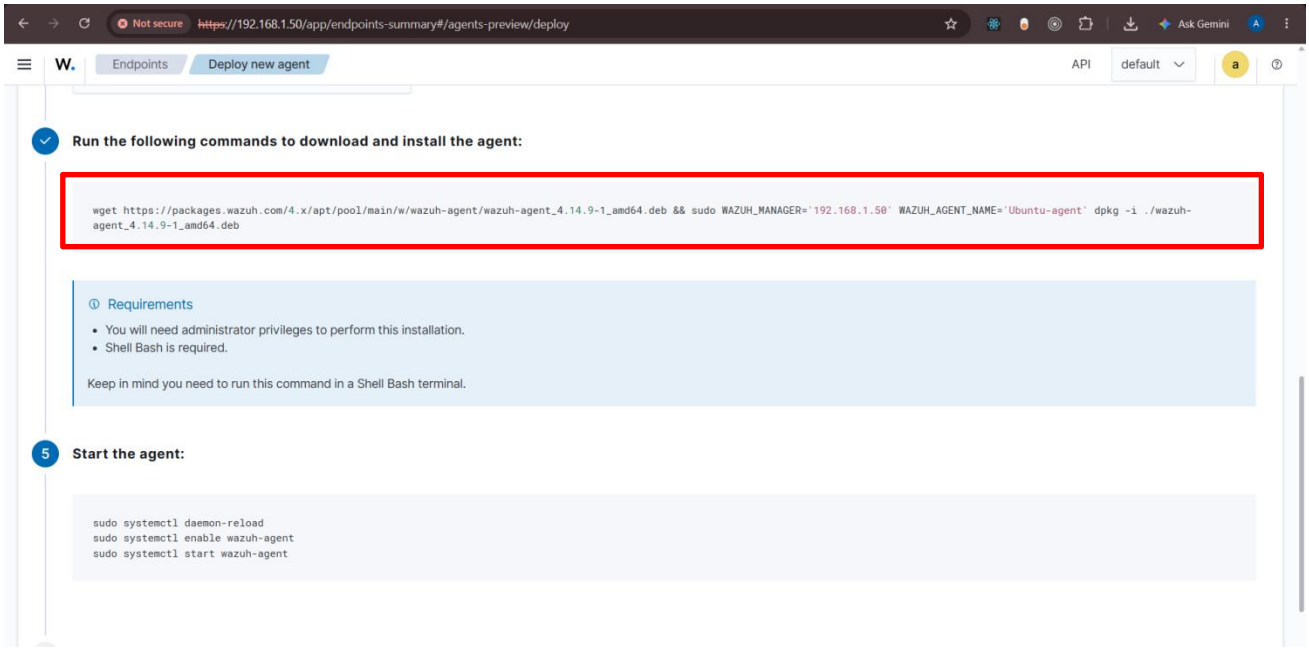
5.2) Installation of Wazuh Agent on Linux



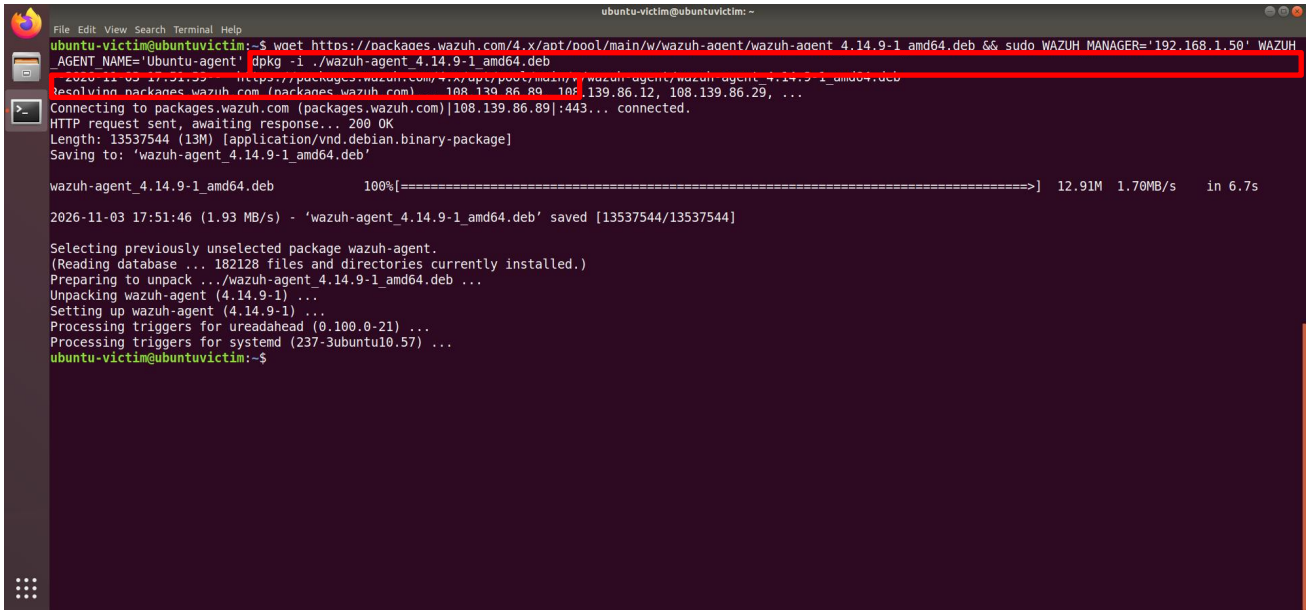
The above screenshot shows the Deploy new agent option being selected in the Wazuh dashboard to begin configuring a Linux endpoint.



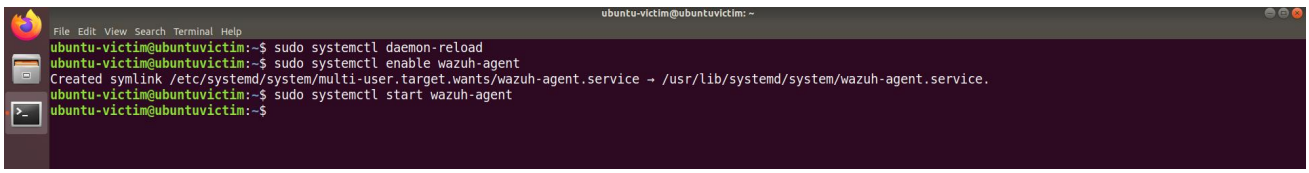
The above screenshot illustrates the Wazuh agent deployment configuration for an Ubuntu endpoint. Linux is selected as the operating system, and the Wazuh server's IP address is specified.



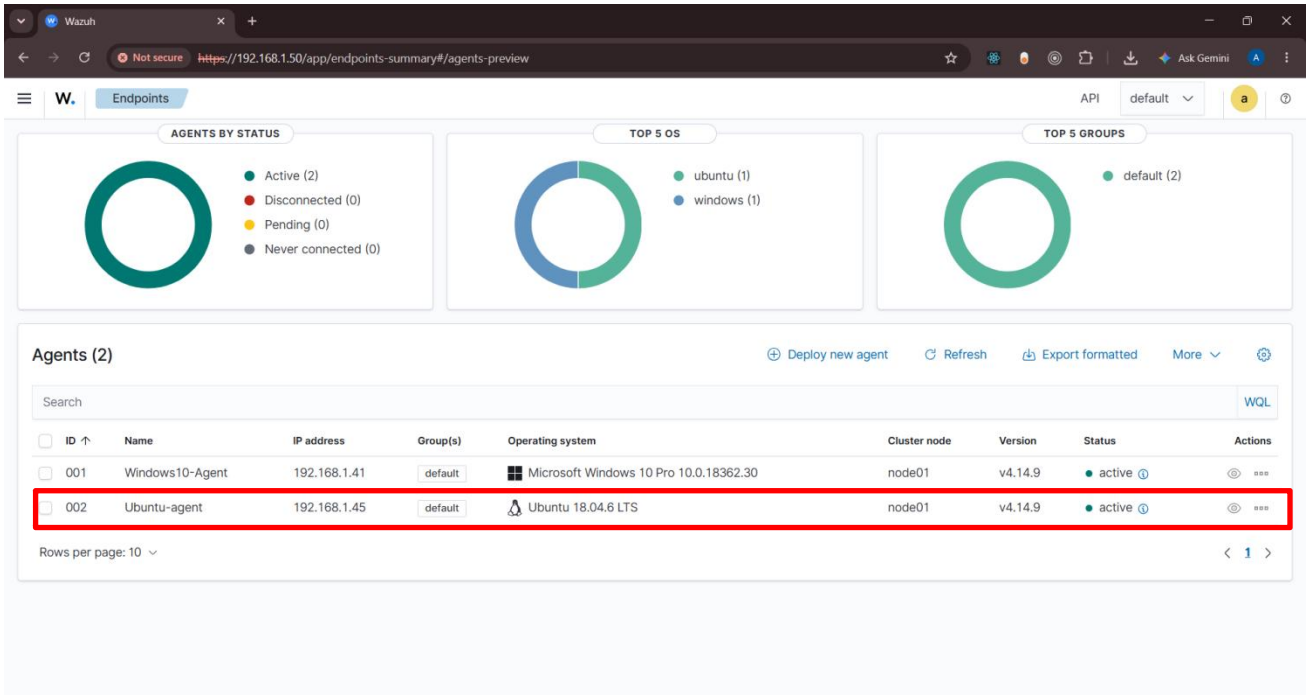
The above screenshot displays the generated installation command for deploying the Wazuh agent on the Ubuntu endpoint.



The above screenshot demonstrates the execution of the Wazuh agent installation command in the Ubuntu terminal.



The above screenshot shows the sequential execution of the commands required to configure and start the Wazuh agent on Ubuntu.



The above screenshot shows the Ubuntu agent appearing in the Wazuh dashboard after installation, confirming that the endpoint has been registered with the monitoring server.

6) Conclusion

This task provided an understanding of Security Information and Event Management (SIEM), its role in centralized log management, threat detection, and incident response. The practical work involved installing and accessing the Wazuh SIEM platform, configuring a static IP address for the Ubuntu server, and deploying Wazuh agents on Windows 10 and Ubuntu endpoints.

Through this exercise, I gained practical experience in connecting monitored endpoints to a centralized security monitoring platform and understanding how endpoint data can be collected for security analysis. The task strengthened my foundational knowledge of SIEM operations and the role of security monitoring tools in identifying and investigating potential security incidents.